FORM TP 2018014

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CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours 15 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE sections and TWELVE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Code is to be written in Pascal.
5. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
6. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION I

THEORY – 60 marks

Answer ALL questions.

1. (a) State whether EACH of the following devices is an input or output devices.

Plotter .....

Joystick .....

OCR .....

Printer .....

Speaker .....

Microphone .....

(6 marks)

(b) State ONE device which could be used for EACH of the following activities:

(i) Participants in an online meeting sharing video to see each other

.....

(ii) A bank teller processing cheques

.....

(iii) A cashier checking out items being purchased in a supermarket

.....

(iv) Electronic marking of 10 000 multiple choice sheets

.....

(4 marks)

Total 10 marks

2. (a) Sharon is at a workshop and decides to join a teleconference with her staff. Which audio output device would you recommend that she use? Give a reason for your recommendation.

Audio output device

.....

Reason

.....

.....

.....

(2 marks)

(b) For EACH of the following, suggest ONE scenario (and provide a reason), in which it would be more suitable to use

(i) a dot matrix as opposed to a laser printer

.....

.....

.....

.....

(2 marks)

(ii) a plotter as opposed to a printer

.....

.....

.....

.....

(2 marks)

(iii) a magnetic tape as opposed to a flash (USB) drive

.....

.....

.....

.....

(2 marks)

(iv) a joystick as opposed to a mouse.

.....

.....

.....

.....

(2 marks)

Total 10 marks

3. (a) Convert 42_{10} to base two.

(3 marks)

(b) Determine the decimal equivalent of 1001_2.

(2 marks)

(c) Write a representation for the positive sign of a binary-coded decimal (BCD).

.....

(1 mark)

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(d) If the ASCII equivalent of the letter P is 1010000, determine the ASCII equivalent for the letter U.

(4 marks)

Total 10 marks

4. (a) At Magnet Defence Systems Ltd there is a 24-hour production system in place. Timely reports must be constantly produced and sent to the engineers for immediate action. A decision must be made on the choice of processing modes to be used: batch, real-time or time-sharing.

(i) Explain which processing mode is MOST suitable for the production of the reports.

.....

.....

.....

.....

(2 marks)

(ii) Describe TWO situations in which one mode, other than the one explained in (a) (i), would be MOST suitable.

.....

.....

.....

.....

(2 marks)

(b) Study the laptop specifications below and answer the questions which follow.

Windows 10 Home 64
Intel® Core ™ i7-6560U Processor
Beautiful 33.8 cm (13.3″) Diagonal Quad-HD OLED (2560 ×1440) Touchscreen display
8 GB RAM and 512 GB PCIe® SSD Storage
500 GB SATA Hard Drive 720 RPM
Intel® IRIS™ Graphics; Audio by Bang & Olufsen
True Vision Full-HD Webcam
Cortana Optimized Microphone; QWERTY Keyboard

State

(i) the primary memory capacity

.....

(ii) the type of hard drive

.....

(iii) the operating system

.....

(iv) the hardware interface

.....

(v) TWO input devices.

.....

(6 marks)

Total 10 marks

5. (a) The following is a list of terms associated with the use of computer systems to facilitate a variety of secondary storage media and devices.

Broadcast, Computer fraud, Extranet, Half-duplex, Simplex, Intranet, Propaganda, Wifi, Point-to-point, Software piracy

Select the appropriate term **from the list** to match EACH of the following situations:

(i) High-speed wireless access to a network or the Internet

.....

(ii) Direct line between two computers

.....

(iii) Data can travel in two directions but only one way at a time

.....

(iv) A private network that belongs to an organization

.....

(v) Unauthorized distribution of computer programs

.....

(vi) Use of computer systems to distribute information

.....

(6 marks)

(b) State ONE advantage and ONE disadvantage of using EACH of the following devices:

(i) DVD

Advantage .....

.....

Disadvantage .....

.....

(ii) External hard drive

Advantage .....

.....

Disadvantage .....

.....

(4 marks)

Total 10 marks

6. (a) The DIGITS company collects personal data from customers which it stores in filing cabinets or on computers.

(i) State TWO possible effects of improper storage of data.

.....

.....

.....

.....

(2 marks)

(ii) For EACH of the following storage locations, state TWO measures that can be used to protect the data:

Filing cabinets .....

.....

Computer systems .....

.....

(4 marks)

(b) Outline when EACH of the following procedures would MOST likely be applied to the company's data:

Recovery .....

.....

.....

(2 marks)

Archiving .....

.....

(1 mark)

Backing up .....

.....

(1 mark)

Total 10 marks

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SECTION II

PRODUCTIVITY TOOLS – 15 marks

Answer ALL questions.

7. The following is an excerpt from an advisory that was sent out to shoppers warning them of the dangers of shopping online during the Christmas season.

CARCIRT-Ad-2017.002v_1 (ALL)

First Release: November 1, 2017

ADVISORY [CHRISTMAS] Caribbean CIRT Christmas Advisory

Important information from CARIBBEAN CIRT

1/11/2017 12:40 PM EST

The Christmas season is one of the times of the year for major shopping. Online shopping is becoming increasingly popular and we are being bombarded with offers of all types. Online shopping is great but shoppers are encouraged to take certain precautions especially when using their credit cards.

(a) (i) State TWO text formatting features, other than bold and underline, used in the document.

.....

(1 mark)

(ii) State the line justification used in the first two lines of the document.

.....

(1 mark)

(iii) What word processing feature will allow the first two lines of this advisory to be repeated on every page of the document?

.....

(1 mark)

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(b) The committee wants to personalize the advisory for distribution to its workers without having to type each worker's name into the advisory.

(i) Name the word processing feature that is MOST suitable to achieve this.

.....

(1 mark)

(ii) List THREE fields that should be inserted into the advisory.

.....

.....

(3 marks)

(iii) Insert the THREE fields listed in (b) (ii) on the advisory **on page 14.**

(1 mark)

Total 8 marks

8. The director of Emrock is creating a budget for the company's small response division, and produces the following spreadsheet which has been sorted.

	A	B	C	D
1	**Items**	**Quantity**	**Cost per Item**	**Total Cost**
2	Digital Forensic Tool Kit	1	3000	
3	Write Blocker Kit	2	1699	
4	Mac Desktop	1	1100	
5	Desktop Workstation	4	900	
6	Wireless Router	2	500	
7	Computer Screens 27″ and 34″	4	400	
8	Response Field Kit	2	300	

(a) Write the formula to be inserted into cell D2 to calculate the total cost of the digital forensic tool kit.

.....

(3 marks)

(b) Outline how to insert the formula in (a) into cell D3 to D8 in order to calculate the totals of the remaining items without retyping the formula.

.....

.....

(2 marks)

(c) State

(i) the field that was used to sort the spreadsheet

.....

(ii) the order used to sort the field.

.....

(2 marks)

Total 7 marks

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SECTION III

PROBLEM-SOLVING AND PROGRAMMING – 45 marks

Answer ALL questions.

9. Consider the following algorithm.

```
INPUT WEIGHT, YEAR
IF (WEIGHT < 180.0) AND (YEAR = 1990)
THEN PRINT "PERFECT"
ELSE PRINT "NOT FIT"
ENDIF
```

(a) (i) Explain the purpose of the program.

.....

.....

.....

(2 marks)

(ii) State the data type of the variable WEIGHT.

.....

(1 mark)

(b) State TWO examples of data that would NOT be suitable to test the pseudocode.

Example 1 .....

Example 2 .....

(2 marks)

(c) Complete the following trace table to show the output from the two algorithms.

WEIGHT	YEAR	IF (WEIGHT< 180) AND (YEAR = 1990) THEN PRINT "PERFECT"	IF (WEIGHT< 180) OR (YEAR = 1990) THEN PRINT "PERFECT"
150	1990		
130	1998		
180	1990		
200	1998		

(5 marks)

Total 10 marks

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10. Consider the following pseudocode for a program.

```
Initialize count to zero
Initialize temp to zero
While temp is not equal to –1
        Input temp
        Output 'temp accepted'
        If (temp = –1) then
            Output 'End of test'
        Else add 1 to count
Output 'Exit'
```

In the space **on page 19**, draw a flow chart for the program.

(15 marks)

Total 15 marks

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11. Consider the following one-dimensional array named CLASS which is used to store the number of students in Class 1 to Class 4.

Class	45	54	63	76
Index	1	2	3	4

(a) State

(i) the number of children in Class 1

.....

(ii) the class that has 54 children.

.....

(2 marks)

(b) Write Pascal code for EACH of the following:

(i) Input the number 63 into Class 3.

.....

.....

.....

.....

.....

(4 marks)

(ii) Add the number of children in the four classes and store the total in a variable named TOTAL.

.....

.....

.....

.....

.....

(4 marks)

Total 10 marks

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12. Each of the following programming statements in the table below is incorrect. Rewrite EACH statement using correct Pascal code.

	Incorrect Programming Statement	Correct Programming Statement
(i)	`Cost: array 1, 8 of real;`	
(ii)	`While (Week < 5)` `  Week = Week + 1;`	
(iii)	`Const` `  INTEREST < 25%;`	

(iv)	`Repeat`
	`  Print Ans;`
	`  Ans : Ans + 1;`
	`Until Ans = 4 or 5;`	

(10 marks)

Total 10 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.